

MALIK JIRASEK

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Institute for Theoretical Physics
University of Tübingen, Germany
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EDUCATION

Oct 2025 – Present

Master of Science in Physics, University of Tübingen, Germany

Thesis: "Keldysh-field-theory approach to the quantum Fisher information"

Supervisor: Igor Lesanovsky

Aug 2020 – Jul 2025

Bachelor of Science in Physics, University of Tübingen, Germany

Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)

Thesis: "Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"

Supervisor: Igor Lesanovsky

RESEARCH EXPERIENCE

Nov 2025 – Present

Master of Science Thesis Research, University of Tübingen, Germany

- Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.
- Demonstrating genuine quantum advantage in driven-dissipative many-body sensor architectures.

Aug 2025 – Present

Working Student, Bosch eBike Systems, Germany

- Implementing a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.
- Collaborating with the engineering team to integrate the model into the production workflow.

Oct 2024 – Jul 2025

Bachelor of Science Thesis Research, University of Tübingen, Germany

- Demonstrated sub-Heisenberg scaling of precision in optical-phase estimation using a boundary time crystal as a light source.
- Designed and numerically benchmarked realistic measurement protocols with QuTiP simulations.
- Results incorporated into a manuscript submitted to *Physical Review Letters* (under review).

Aug 2024 – Oct 2024

Research Intern, Humboldt University of Berlin, Germany

- Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving 10^{-8} relative precision in ground state energies.

PUBLICATIONS

In preparation

Malik Jirasek, Igor Lesanovsky, and Viktoria Noel, "Keldysh-field-theory approach to the quantum Fisher information". *In preparation*.

Preprint / under review

Malik Jirasek, Igor Lesanovsky, and Albert Cabot, "The Boundary Time Crystal as a light source for quantum enhanced sensing beyond the Heisenberg Limit". *arXiv: 2511.23416 [quant-ph]*, 2025. Submitted to *Physical Review Letters* (under review). *arXiv:2511.23416*

PRESENTATIONS & CONFERENCES

06 Mar 2026

Contributed Talk at DPG Spring Meeting SAMOP, Mainz

TEACHING EXPERIENCE

<i>WS 2024/25</i>	Advanced Quantum Mechanics tutor (University of Tübingen)
<i>SS 2024</i>	Quantum Mechanics tutor (University of Tübingen)
<i>WS 2023/24</i>	Analytical Mechanics tutor (University of Tübingen)
<i>SS 2023</i>	Physics 2 tutor (University of Tübingen)
<i>WS 2022/23</i>	Physics 1 tutor (University of Tübingen)
<i>WS 2021/22 - WS 2024/25</i>	Physics Lab tutor (University of Tübingen)

AWARDS & SCHOLARSHIPS

<i>2021 & 2023</i>	Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)
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SKILLS & METHODS

<i>Programming</i>	Python (TensorFlow, QuTiP), Fortran, MATLAB (Simulink)
<i>Simulation & Modeling</i>	Lindblad formalism, Keldysh formalism, stochastic calculus
<i>Tools</i>	Linux, Git, LaTeX
<i>Languages</i>	German (native), English (full professional proficiency), French (beginner)

SERVICE & LEADERSHIP

<i>Oct 2026</i>	Co-organizer, Workshop "Nonequilibrium methods in driven-dissipative many-body systems"
<i>2026</i>	Student representative, faculty appointment committee (ML in physics)
<i>2022 - 2025</i>	Treasurer & member, physics student council

REFERENCES

Available on request.